LESSON 6.11

# Fraction Division Problem Solving

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can solve real-world problems by deciding whether the question asks how many groups fit or how much is in one group, writing a fraction division equation, and checking the solution by multiplying.

**WHY THIS MATTERS** Agent Torres is closing the biggest case of the year. She has 2/3 of a gallon of fingerprint dust left and each test uses 1/6 of a gallon. Before the evidence expires, she needs to figure out exactly how many tests she can run. Can you write the equation and solve it?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Model** *Spanish: Modelo* | A picture or math way to show a problem so you can solve it. | A bar model showing 3/4 split into 1/8-size pieces, or the equation 3/4 ÷ 1/8 = 6 |
| **Equation** *Spanish: Ecuación* | A math sentence with an equal sign showing both sides are the same. | 3/4 ÷ 1/8 = 6 — the left side (division) equals the right side (answer) |
| **Solution** *Spanish: Solución* | A number that makes the equation or inequality true. | In 3/4 ÷ 1/8 = 6, the solution is 6 portions |
| **Reasonableness** *Spanish: Razonabilidad* | Checking if your answer makes sense. | 3 ÷ 1/4 = 12. Is 12 reasonable? Yes, because 1/4 is small so many pieces fit into 3. |
| **Inverse operations** *Spanish: Operaciones inversas* | Two math actions that undo each other, like × and ÷. | If 3/4 ÷ 1/8 = 6, then 6 × 1/8 = 6/8 = 3/4. Multiplication checks division. |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Agent Torres has 2/3 gallon of dust and each test uses 1/6 gallon. I ask: how many tests fit?**

1. The total 2/3 is the dividend. The size of each test 1/6 is the divisor. My equation is 2/3 ÷ 1/6.
2. In sixths, 2/3 = 4/6, and 4 sixths ÷ 1 sixth = 4. Multiplying by the reciprocal gives the same answer: 2/3 × 6/1 = 12/3 = 4.
3. So she can run 4 tests. I check by multiplying back: 4 × 1/6 = 4/6 = 2/3. It matches the total, so the answer is reasonable.

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**A ribbon is 3/4 yard and each bow uses 1/4 yard. The total 3/4 is the dividend; 1/4 is the divisor.**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The total amount is the _____ because it is being split into groups. The size of each group is the _____ because we are dividing by that amount.

**Word bank** Model | Equation | Solution | Reasonableness | Inverse operations | dividend | divisor | total | group | split | equation

WATCH OUT!

## A common mistake

A common mistake in Fraction Division Problem Solving is reversing which fraction is the dividend and which is the divisor when setting up the equation from a word problem. For example: a pitcher holds 3/4 cup of punch, and each serving is 1/8 cup

LESSON 6.11

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**Level 0** · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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## PART 1 Understand the idea

models and meaning

**1** **Write the name of the correct group on each line.**

Sort each word problem: should you DIVIDE or MULTIPLY to solve it?

**Groups:** Division Problem | Multiplication Problem

_____ How many 1/3-cup scoops fit into 2 cups? (2 ÷ 1/3)

_____ A ribbon is 3/4 yard. Each piece is 1/8 yard. How many pieces? (3/4 ÷ 1/8)

_____ You run 1/2 mile, 3 times. How far total? (1/2 × 3)

_____ A recipe uses 2/3 cup. You make half a recipe. How much? (2/3 × 1/2)

STEPS TO TRY
1. Re-read each story: does it ask 'how many fit?' or 'how much is a part of the amount?'
2. 'How many groups or pieces fit' → division. 'A fraction OF an amount' → multiplication.
3. First card: 1/3-cup scoops filling 2 cups asks how many scoops FIT — decide which lock that clue belongs in.

## PART 2 Apply it

real-world problems

**2** **Circle the letter of the best answer. Show how you know.**

A ribbon is 4/5 of a yard long. Each bow uses 1/10 of a yard. Which equation finds how many bows can be made?

(A) 1/10 ÷ 4/5

(B) 4/5 ÷ 1/10

(C) 4/5 × 1/10

(D) 4/5 + 1/10

**STEPS TO TRY**

1. Name the roles: 4/5 yard is ALL the ribbon; 1/10 yard is what one bow uses.
2. 'How many bows' means total ÷ group size — and the total always comes before the ÷ sign.
3. Decide which number is being cut up. Put that one first, then check each choice's order and operation.

**SHOW YOUR WORK**

**3** **Circle the letter of the best answer. Show how you know.**

A detective has 1/2 gallon of solution. Each test uses 1/8 gallon. How many tests can be run?

(A) 1/16

(B) 1/4

(C) 4

(D) 16

**STEPS TO TRY**

1. The total is 1/2 gallon; each test drains 1/8 gallon — you're counting tests.
2. Rewrite 1/2 as eighths so both amounts match, or the reciprocal shortcut with 8/1.
3. Multiply top and bottom of 1/2 by 4 to get eighths, then count how many 1/8-tests that covers.

**Rewrite each fraction** _____

**Work it out** _____

**Simplify** _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

**4** **Circle the letter of the best answer. Show how you know.**

A trail is 3/4 mile long. A jogger runs laps of 1/4 mile each. How many laps does the jogger complete?

(A) 1/3 lap
(B) 3 laps
(C) 3/16 lap
(D) 4 laps

**STEPS TO TRY**

1. The trail is 3/4 mile total, and every lap covers exactly 1/4 mile.
2. Both numbers are fourths — count how many 1/4s stack up to reach 3/4.
3. Stack laps one at a time: 1/4, then 2/4, then… stop when you reach 3/4 and count your laps.

**Rewrite each fraction** _____

**Work it out** _____

**Simplify** _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

**EXPLAIN YOUR THINKING**

**How do you decide which value is the dividend and which is the divisor in a word problem?**

**SENTENCE STARTER** The total amount is the ___ because it is being split into groups. The size of each group is the ___ because we are dividing by that amount.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help